COINPICKS · RESEARCH & EDUCATION

Bitcoin Technical Analysis
Quant Pillar Report

The Tape Itself — Measured, Scored, and Mapped to the Pillar Standard

Machine-computed from 16 years of daily price data (5,837 days, 2010–2026) through a 7-indicator scoring stack, halving-cycle analogs, and a power-law valuation model. Every number is reproducible from the scripts in the BTC-TA folder. ***Scores are the model's draft — awaiting my stamp.***

COINPICKS QUANT PILLAR REPORT · THE QUANTITATIVE SLEEVE OF MY THESIS

Data through July 11, 2026 (BTC $64,128) · Pipeline re-run & verified July 11, 2026 · Not Financial Advice

The Scores — July 11 Model Draft

Model draft computed July 11, 2026 — same standard as every qualitative pillar. Adjust any P and re-stamp. (July 10 draft was 39/38/44/67/70 — the 6-month window firmed as the cycle analog flipped.)

WINDOW	DRAFT P	QUALITY	IMPACT	SIGNAL	CALL
1 month	**39**	0.75	8	**−1.3**	Mild Bear — broken tape, first repair signal holding
3 months	**38**	0.70	8	**−1.3**	Mild Bear — the final-flush window
6 months	**47**	0.55	8	**−0.3**	Neutral (borderline Mild Bear; exact −0.26)
1 year	**67**	0.60	8	**+1.6**	Bull — the cycle turn
3 years	**70**	0.50	8	**+1.6**	Bull — value + cycle

Impact: 8 (the tape aggregates every driver, but adds no causal force of its own) · **Formula:** Signal = ((P − 50) ÷ 50) × Quality × Impact · **Honesty cap:** P = 50 + 25 × (window score ÷ 100) — pure TA never claims odds more than 25 points from a coin flip, because its measured out-of-sample edge is modest. Math machine-verified.

THE THREE INPUTS, IN PLAIN ENGLISH

P — the odds the tape is *good for Bitcoin* by then. Near-term low (weekly composite −48/100, all seven indicators still bearish — though price is holding above the 200-week average); long-term high (at this exact point in prior cycles, the next 12 months were positive 3-for-3, and price sits 52% below its power-law fair value).

Quality — how trustworthy the evidence is. Highest near-term (835 weekly bars, unanimous indicators), lowest at 3 years (only 3 completed cycles to learn from, and this cycle is already breaking the old pattern).

Impact — 8 out of 10, now the uniform value shared by every pillar; the TA pillar carries an equal 20% weight in the master, same as each qualitative pillar. Price is the thing itself — but a mirror, not an engine.

The Claim

The thesis, and where the numbers differ from the mood

The crowd: −49% from the all-time high, a lower low on June 30, a weekly close under the 200-week average as recently as June 28 — "the cycle is dead, get out." (The bounce to ~$64.1K, now three sessions straight through the July 8–9 Iran escalation, is the counter-evidence the crowd is ignoring.)

The model: the short-term tape is still broken (P 39/38 for 1–3 months — this is not yet a dip-buying report), and the 6-month window is now a coin flip (P 47) rather than mildly bearish. But the same data says P 67 at one year: every prior cycle at this exact month looked just as dead and was higher 12 months later, price is deep below fair value, and the froth that precedes real crashes is absent.

The edge: the crowd reads one clock (the tape). The model reads three — tape (bearish), cycle clock (bottoming window: prior cycles bottomed at months 26–30; we're at month 27), and valuation (52% below the power-law fair line, under the −1σ "cheap" band). Two of the three clocks are already on the bull's side; the tape is the last one, and it always turns latest.

The Facts — The Tape

Computed July 11, 2026 from Bitstamp daily data (BTC $64,128, daily mid-session). Current weekly/monthly/quarterly bars are still forming — scores use them as-is. Quantified: on completed bars only (the June 28 weekly / June 30 monthly / June 30 quarterly closes), the *weekly composite is −51 and monthly −43; the current quarterly bar holds just 11 of ~92 days.*

- **Weekly composite −48/100 (Moderate Bearish), 7-of-7 indicators agree.** Price is below the 20-week ($69,939, −8%) and 50-week ($87,504, −27%) averages but holding **above the 200-week ($62,872, +2%)**; RSI 38.9; MACD line below zero (−5,828) — negative momentum, though the histogram keeps improving (−344 → −210 → −201 over three weeks); %B 0.27; OBV still shows distribution.
- **The first repair signal is holding.** The June 28 bar delivered the first weekly close below the 200-week line since October 2023; the July 9–11 bounce has kept price above it for a third straight session ($64,128 vs $62,872 weekly-basis, +2.0%). Still unconfirmed until Sunday's weekly close — the 20-week average (~$69.9K) is the confirmation that matters.
- **Monthly composite −30/100 (Slight Bearish), 5-of-7 agree.** Below the 20-month average ($89,027, −28%), still above the 50-month ($59,905, +7%); RSI 43.9; monthly MACD histogram −6,038 — deeply negative, marginally improved from June's −6,243; %B 0.14 — near the lower band. The monthly rising support trendline is still broken (line at $68,368, price −6.6% under it).
- **Quarterly composite +9/100 (Neutral), agreement 1-of-7** — the long tape still holds its bull structure (above the 20-quarter average, $57,428, +12%) but with falling momentum and heavy quarterly OBV distribution. The weakest, noisiest layer of the stack.
- **Levels that matter:** support $59,930 (weekly + monthly swing), then $49,577. Resistance $70,016, then $74,434, then $81,685. The falling weekly support trendline sits at $64,432 — price is now just −0.5% below it (was −0.7% on July 10); a close above it would be repair signal #2.

★ **June 30 remains the cycle low: $58,526 close, $57,735 intraday on July 1** — the flush tagged the Fib golden pocket ($54,257–$57,802 = 61.8–65% retrace of the $15,479 → $126,272 run) and has now bounced **+9.6% to $64,128** — through a live Iran escalation. **$58,526 is the line in the sand.**

Weekly tape, scored −48/100 with all seven indicators bearish: below the 20- and 50-week averages, holding above the 200-week, broken trendline, MACD regime negative. Machine-annotated from the same data that feeds the scores.

The Facts — The Cycle Clock & The Value Floor

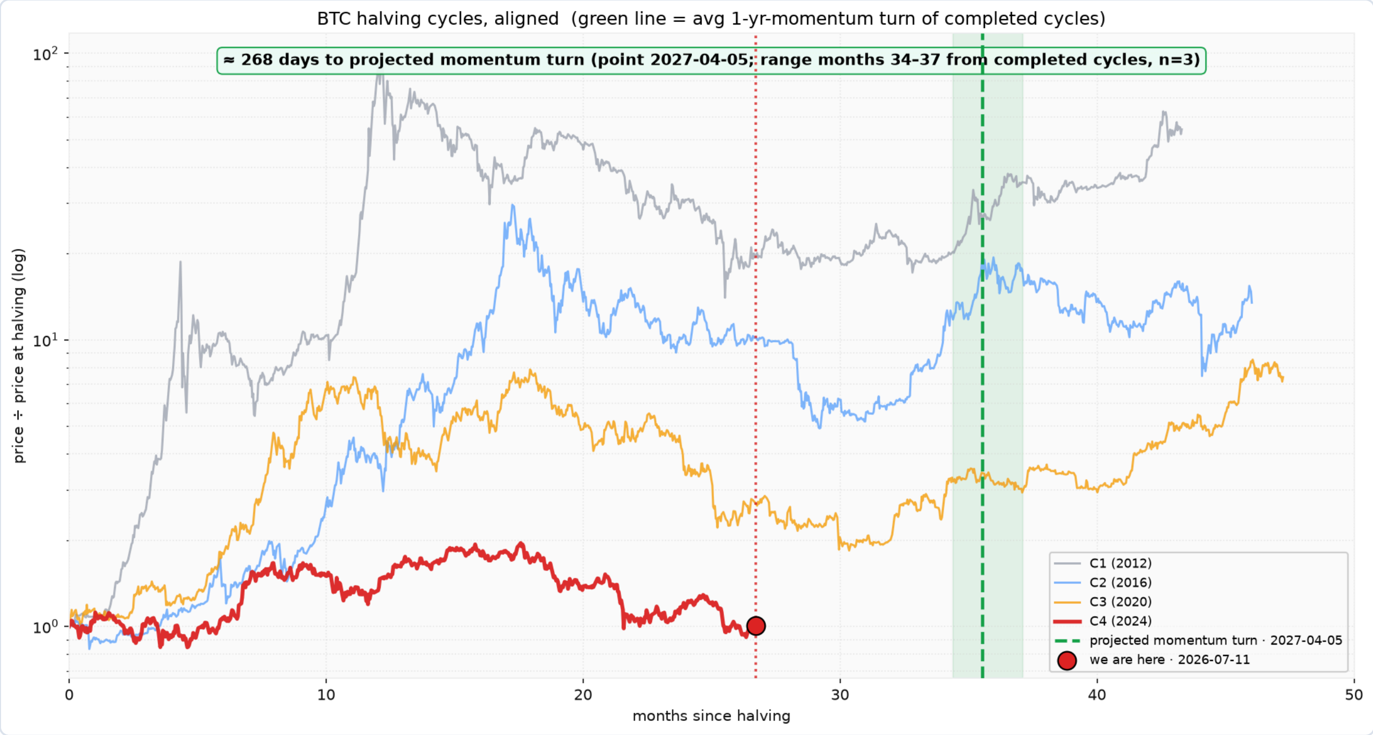
The two slower clocks. Same data, longer lenses.

- **The halving clock says "bottoming window."** The three completed cycles topped 12–18 months after their halving and bottomed at months 26–30 (avg 28). This cycle topped at month 18 (Oct 6, 2025 — $124,728) and printed its low-so-far at month 26 (June 30). **We are at month 26.7; the average-bottom date projects to ~Aug 30, 2026.**
- ★ **The cycle analogs are the spine of the long-window scores — and the 6-month one just flipped again.** Standing at this exact cycle-month in C1/C2/C3: next 1 month positive 1-of-3 (median −5%) · next 3 months positive **0-of-3** (median −11%) · next 6 months positive **1-of-3** (median −2%, was 0-of-3 / median −4% on July 10) · next 12 months positive **3-of-3** (median +27%) · next 36 months positive **3-of-3** (median +530%). Bearish quarter, softening half-year, bullish year — the data's own words. *Fragility disclosure: the 6-month analog has now flipped twice in three days (1-of-3 on Jul 8 → 0-of-3 on Jul 10 → 1-of-3 on Jul 11) purely from the day-count alignment sliding across a 2015 window boundary. With n=3, single windows flip like that; treat the analog levels as soft, the direction as the signal. And know what the 36-month number is: from month ~27 that window reaches month ~63 — it contains the NEXT halving's rally.*
- **1-year momentum turned positive at months 34–37 in every completed cycle** → projects to **~April 2027** (80% window Jan–Jul 2027) for this cycle.
- **The power-law floor:** a log-log fit over the full 16-year series (R^2 0.96) puts fair value at **$135,003**. Fit-window honesty: starting the fit in 2011–2015 instead moves fair value between ~**$114K and $141K** (price 45–56% below fair), and R^2 this high is near-automatic for this kind of regression — the level is soft, but the **"below the −1σ cheap band" verdict holds across every window tested** (−1σ $67,414; −2σ $33,664). Every halving-era cycle

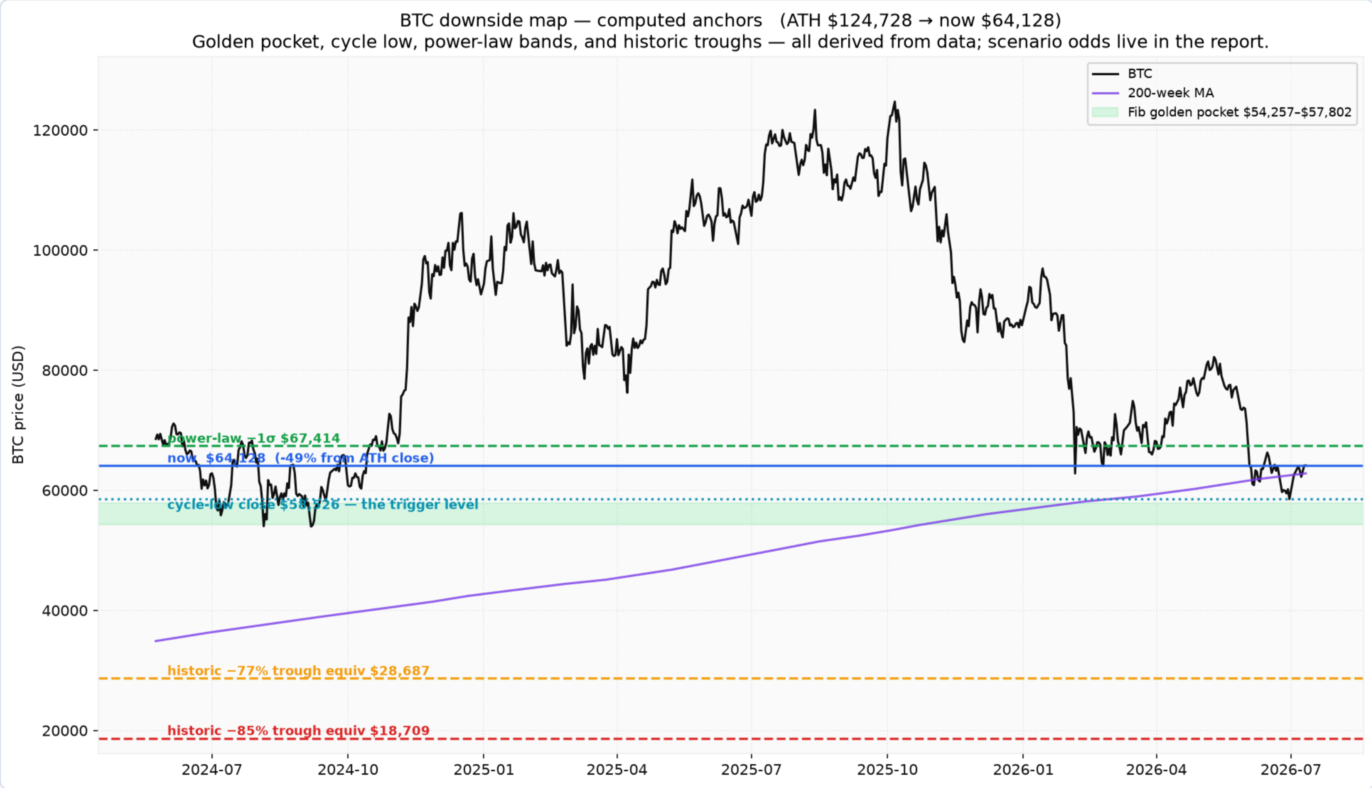
trough was −77% to −85% from ATH ($18.7K–$28.7K equivalent today) — the pre-halving 2011 bear went ~−93% (~$8.7K equivalent), excluded from the range but in the data; this cycle is −53% so far.

- **No froth in the system:** the altcoin-froth index (BTC dominance 77%, ETH/BTC 0.028) reads z = −0.87 — the opposite of the euphoric signature that preceded prior deep crashes. Validated on 16 years of data: high-froth days led to −39% average 12-month drawdowns vs −23% for low-froth days (gap significant, 95% CI excludes zero). *Data lags ~6 weeks (through May 23) — used as a tilt, not a score input.*

★ **What we rejected:** a 5-feature logistic crash model was tested leave-one-cycle-out and **failed** (AUC 0.44 vs 0.58 for froth alone; 0.70 = useful). It is not in this report's numbers. The scores rest only on what survived out-of-sample testing.



All four halving cycles aligned (log scale, price ÷ price-at-halving). Triangles mark where 1-year momentum turned positive in each prior cycle — months 34–37, projecting to ~April 2027 for this one. The red dot is today: month 27, the historical bottoming window.



The downside map, updated after the June 30 flush: the projected $54K–$61K bottom zone was hit ($58,526 close / $57,735 intraday, inside the golden pocket's doorstep) and bounced. Odds re-weighted accordingly: ~40% the low is in, ~35% one more flush to $54–58K, ~17% deeper bear, ~8% tail.

The Prize

Illustrative only — not price targets, not advice

From $64,128: the 20-week average (~$69.9K, trend-repair confirmation) is **+9%**; a retest of the $124,728 ATH is **+94%**; the power-law fair line is **+111%**. The median 12-month analog from this cycle-month is **+27%** (~$81K by mid-2027); the median 36-month analog is +530% — **discount that one hard**, it leans on two regimes (2017 mania, post-COVID ZIRP) that don't repeat on demand.

The Prize doesn't feed the math — it tells you what the lean is worth acting on. Here it says: the payoff for surviving a bearish quarter is a full cycle turn.

The Other Side

The strongest honest case the model is wrong

- **n = 3.** Every long-window claim rests on three completed cycles. Three coin flips landing heads is not a law of nature.
- **This cycle already broke the pattern.** A −53% drawdown at month 26 vs −77/−85% in prior cycles could mean dampened cycles (bullish read) — or it could mean *the bottom isn't in*: a historic-depth trough from this ATH is $18.7K–$28.7K, far below every level in this report.
- **The ETF era changed the machine.** Prior cycles had no spot-ETF flow channel, no CME basis trade at this scale, no corporate treasuries. Analogs may simply not transfer.
- **The power law is a curve fit, not a promise.** R^2 0.96 in-sample says nothing about a structural break — a fit like this "said cheap" through most of 2022's decline too. It bounds scenarios; it doesn't time anything.

- **The tape's edge is small by construction.** The honesty cap exists because measured predictive validity is modest (froth AUC 0.58). If you removed the cap, this report would be louder — and less true.

The Triggers

Price and dated events — some prove the read right, some prove it wrong

WHEN / LEVEL	EVENT	MEANS
$58,526	Daily close below the June 30 low	Flush scenario live — golden pocket $54.3–57.8K is the target; re-rate 1/3-month down
~$62,900	Weekly close above the 200-week MA ($62,872)	**Holding intra-week for a 3rd straight day (Jul 9–11)** — still needs to hold into Sunday's weekly close
~$69,900	Weekly close above the 20-week MA	Trend repair confirmed — re-rate 1/3-month up
~Aug 30, 2026	Historical average-bottom month (28) passes	A higher low after this date + improving monthly momentum = basing confirmed
Jan–Jul 2027	Projected 1-yr momentum turn window (point: ~Apr 2027)	Arriving without new lows = cycle thesis on track; re-rate 1-year up
$50,000	Weekly close below	−60% from ATH; the dampened-cycle story is failing — see the flip line

The flip line: a decisive weekly close below **$50,000** breaks the base case and the golden pocket both — treat the dampened-cycle thesis as broken and put the historic trough zone ($18.7K–$28.7K) back on the table. Above it, a bearish quarter is the plan, not a surprise.

My Decision

◷ **MODEL DRAFT — JULY 11, 2026 · AWAITING MY STAMP**

Draft P: 39 / 38 / 47 / 67 / 70 · Quality: 0.75 / 0.70 / 0.55 / 0.60 / 0.50 · Impact: 8

This is the quantitative sleeve of the thesis — the same five windows, same formula, same table as every qualitative pillar, but computed instead of judged. The near-term read is the honest one: the tape is broken (first repair signal holding for a third day, still unconfirmed) and the analog quarter is the worst window in the set. The 6-month window is now a genuine coin flip rather than mildly bearish. The long read is where the quant leans bullish: cycle clock + valuation + absent froth.

These numbers enter the master blend as drafted. Override any P and re-stamp — the blend recomputes.

What Changed Since Last Check

Pipeline re-run July 11, 2026 (previous run July 10, 2026)

★ **The bounce extended a third day: $63,832 → $64,128 (+0.5%), still through the aftermath of the July 8–9 Iran escalation.** June 30's $58,526 remains the cycle low; price is now +9.6% off it.

- **Repair signal #1 is holding, not yet confirmed:** price has stayed above the 200-week average (weekly-basis $62,872) for a third straight day — still unconfirmed until Sunday's weekly close. The falling weekly trendline ($64,432) is now just 0.5% overhead — repair signal #2 if taken.
- **Composites roughly flat, still bearish:** weekly −49 → **−48** (still 7-of-7), monthly −30 → **−30** (5-of-7), quarterly +9 → **+9**.
- ★ **Draft P moved 39/38/44/67/70 → 39/38/47/67/70.** The 6-month P rose 3 points as the cycle analog flipped back: 0-of-3 positive → 1-of-3 positive (median −4% → −2%) — the same n=3 alignment fragility that dropped it 2 points on July 10 pushed it back up today. Treat the direction, not the exact level, as the signal.
- **Data refreshed:** Bitstamp daily through July 11, 2026 (5,838 days); Coin Metrics altcoin set still through May 23 (community data lags ~6 weeks — froth is a tilt, not a score input).

Method & Sources

Everything is reproducible — the scripts and data live in the BTC-TA folder. Run order below.

- **Price data:** Bitstamp OHLC API (daily, 2011–2026, no key) via fetch_btc_history.py · Coin Metrics community reference price (2010–2011 close-only) via build_master_history.py — 5,835 days total, each row tagged by source.
- **Indicator stack:** indicators.py — SMA 20/50/200, EMA 12/26, RSI-14 (Wilder), MACD 12/26/9, Bollinger 20/2, OBV — on weekly (W-SUN), monthly, quarterly bars. levels.py — swing-cluster S/R + least-squares trendlines.
- **Scoring:** score.py — each indicator → conviction in [−100, +100] (bounded inputs mapped linearly, unbounded squashed with tanh); composite = fixed weights (MA .20, RSI .15, MACD .15, BB .15, S/R .15, OBV .10, TL .10); agreement = indicators sharing the composite's sign.
- **Cycle & value:** cycle_analysis.py — halving-aligned tops/bottoms/momentum turns, projections from C1–C3 · downside.py / downside_map.py — power-law bands (R^2 0.96) + historical troughs + 6-method bottom-zone synthesis · test_altcoin_froth.py — froth quartiles vs forward drawdowns, block-bootstrap CI.
- **The mapping (new):** pillar_mapping.py — window scores from tape composites + cycle analogs + power-law value; P = 50 + 25×(S/100); Quality rubric per window; writes data/pillar_scores.json for the master blend.
- **Run order:** fetch_btc_history → build_master_history → fetch_altcoin_data → indicators → levels → score → charts → cycle_analysis → downside → downside_map → pillar_mapping.
- **Honesty notes:** current weekly/monthly/quarterly bars are partial as of July 11 (complete-bar composites: weekly −51 / monthly −43 / quarterly +7.7) · the multi-signal crash model failed out-of-sample (AUC 0.44) and is excluded · the AUC figures are the pipeline's own attestation · froth data lags ~6 weeks · n=3 on all cycle claims, and analog windows can cross into the next cycle — the 6-month analog has flipped twice in three days on pure alignment drift, printed in the Facts · the 2024 halving is dated Apr 19 (US ET; block 840,000 hit Apr 20 00:09 UTC — 0.03 cycle-month effect) · S/R levels are algorithmic swing-clusters with 1–2 touches each · the bottom-odds split (40/35/17/8) is a judgment synthesis, not a model output.